Mude Khusinadha Naik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.09
Sainik School Kalikiri	Class XII (CBSE)	2023	87.20%
Sainik School Kalikiri	Class X (CBSE)	2021	92.40%

EXPERIENCE

Kotak Mahindra Bank | Software Engineer Intern

- Developed backend services for **BPC (Bill Payment & Collection)**, a **Spring Boot**-based system automating utility bill fetching and payment processing, integrating with **Billdesk** for **UPI/NEFT/RTGS** transactions.
- Built a **CRON-orchestrated batch scheduler** with automatic stall detection and self-healing recovery, processing **150+ batches/hour** across the payment pipeline without manual intervention.
- Designed a parameterized **JSON notification template system** and a **REST API** for manual refund file uploads to **AWS S3**, handling edge-case payment failures requiring operations review.
- Onboarded enterprise banking data from **Oracle, SQL Server, and PostgreSQL** into a unified analytics platform via a **CDC pipeline (GoldenGate Kafka Flink StarRocks/Iceberg)**, deploying **8 production tables** and validating **40+ in UAT**, and authored a custom **Claude Code** skill that streamlined the table onboarding workflow for the wider team.

PROJECTS

Counter-Strike Match Outcome Predictor | *Python · scikit-learn · pandas · Monte Carlo simulation · Markov chains*

- Built an end-to-end **CS2/CS:GO match prediction engine** in **Python**, trained on **1,990 real HLTV professional matches** across **59 teams** and **8 maps**.
- Engineered **leakage safe, time decayed team features** (win rate, per-map form, per-side form, momentum, win streaks) with **Laplace smoothing**, and tuned hyperparameters using **chronological cross validation** to avoid future data leaks.
- Trained a **logistic regression model** to blend the features, reaching **61.9% test accuracy** against a **57.5% rank-only baseline**, and ran an **ablation study** showing the **rank + win rate feature pair** alone hit **62.9%**.
- Modeled **team economy states as a 4-state Markov chain** (Eco / Force / Half_Buy / Full_Buy) and built a **dynamic-programming live win-probability engine** over (score, economy) states for **round-by-round match forecasting**.
- Ran **5,000 iteration Monte Carlo tournament simulations** and a **378,000 match round robin season simulation** across **28 teams**, correctly predicting the champion on the held-out verification scenarios.

Real Time Video Game Object Detection System | *YOLOv8 · PyTorch · OpenCV · Roboflow*

- Sourced and version managed a **10,000 image annotated dataset** via Roboflow, exporting directly into YOLOv8-ready train/validation/test splits to eliminate manual annotation reformatting.
- Trained a **YOLOv8s object detection model** (**11.1M parameters**) to localize 4 in-game classes -- teammates, enemies, dropped spike, planted spike -- from live Valorant gameplay footage.
- Achieved **99.4-99.5% mAP@50** with **99.3-99.6% precision/recall** across validation (1,983 images) and a held-out test set (988 images), with **no drop in test performance**, indicating the model generalizes rather than memorizing training data.
- Benchmarked **real-time inference at ~11ms/frame (89 FPS)** on consumer GPU hardware, and validated results with a **visual comparison pipeline** overlaying predicted bounding boxes against groundtruth annotations.

GitHub Repo Intelligence | *Python · JavaScript · Flask · Chrome Extensions · Groq API · GitHub REST API*

- Engineered an **AI-powered Chrome extension** that injects interactive, context-aware sidebar into **GitHub repository pages** to accelerate developer onboarding.
- Architected a local **Flask backend** that aggregates **GitHub REST API** data (file tree, README, metadata) and prompts **Groq's Llama 3.3**-model to return structured JSON insights, including inferred architecture patterns, deployment targets, and optimal code entry points.
- Built a **related-repository recommendation feature** that constructs topic- and keyword-based **GitHub Search API** queries, ranked by star count, to surface comparable repositories.
- Delivered **sub-3-second end-to-end analysis** for complex, multi-language codebases via **asynchronous data**

fetching.

Digital Noise Filter | *MATLAB · Signal Processing · Adaptive Filters*

- Designed and implemented a **real-time adaptive noise cancellation system** in **MATLAB** using the **NLMS algorithm**. The filter estimates the noise component in a noisy speech signal and subtracts it out on a sample-by-sample basis.
- Built a configurable cascade of **second-order IIR notch filters** to selectively suppress known tonal noise components, such as electrical hum, ahead of the adaptive stage. The pipeline supports both full-adaptive mode and a hybrid notch-plus-adaptive mode.
- Validated performance using **SNR (Signal-to-Noise Ratio)** analysis against ground-truth clean speech, achieving a **20 dB improvement in SNR** after filtering.

TECHNICAL SKILLS

- **Programming Languages :** C, C++, R, Python, Java, JavaScript, HTML, CSS, SQL, MATLAB, Verilog

- **Core Competencies :** Data Structures & Algorithms, Object Oriented Programming, Operating Systems, DBMS, Deep Learning

- **Tools and IDEs :** LTSpice, RStudio, Arduino IDE, VS Code, Neovim, IntelliJ, Claude Code, DBeaver, Postman

- **Frameworks & Technologies :** Flask, Spring Boot, Docker, REST APIs, PyTorch, OpenCV, LLM API Integration

- **Platforms & Version Control**: Git, GitHub, Jupyter Notebook, Linux Bash

ACHIEVEMENTS & EXTRACURRICULAR

- Class representative for the EE23 BTech batch

- Avid problem solver with 500+ submissions on LeetCode, Codeforces - Pupil, and HackerRank

- Committed athlete, member of NSO Volleyball team and active participant in institute sports leagues (Milan, Diesta)

- Esports enthusiast : participated in IVL, Fall guys , CoD Mobile league (2nd position)